Computing Between Models with Residual Coupling

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Abstract

We propose Residual Coupling (RC), a modular architecture for connecting frozen language models through small learned bridge operators that inject corrective updates additively into each model's residual stream, in the same way that residual connections [5] within a single transformer add layer outputs to the stream. No base model weights are modified at any point.

We view frozen transformers as non-linear memorizers whose weights define the boundaries of what the system can know. The bridge projections are linear operators constrained to navigate the structured differences between these memorized representation spaces. Because they are linear, their capacity to represent arbitrary mappings is sharply constrained compared to a non-linear layer. Instead, they learn continuous relational maps between what the frozen models have separately encoded, reaching positions in the relational geometry that neither model has individually represented.

We evaluate RC across four domains and find that it consistently outperforms Mixture-of-Experts routing using the same frozen models. In the medical domain with three coupled models, multi-bilateral coupling reduces perplexity by 80.7% relative to the frozen baseline, compared to 0.5% for MoE. Factual accuracy on TruthfulQA Health improves by up to 9 percentage points over the frozen baseline and by more than 5 percentage points over MoE. New domains are added by training bridges to a new specialist module, and specialists are removed without any retraining. Because no model weights are ever overwritten, catastrophic forgetting of base representations is structurally prohibited.

1. Introduction

The dominant approaches to specializing language models all share a common assumption: that capability resides inside individual models and the task is to configure or select among them. Fine-tuning writes new knowledge into a model's weights. Liu et al. [15] show that it more precisely reactivates memorized content already present from pretraining, recovering verbatim passages from training corpora after targeted fine-tuning. Mixture-of-Experts routing [4, 21] commits each token to a single expert and discards the others. Agentic pipelines pass outputs between models as text, compressing each model's continuous intermediate latent state-space into a discrete token sequence. In all three cases, the internal representations of one model do not directly influence the computations of another.

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There is also a broader architectural contrast worth noting. Progress in language modeling has largely followed a vertical path, scaling individual models in depth and width. RC offers a horizontal alternative: capability is distributed across frozen models of fixed size, and the system grows by adding bridge connections rather than by scaling any individual model.

We propose Residual Coupling (RC), a framework in which two or more frozen models are connected through small learned bridge operators at designated intermediate layers. The bridge at layer ℓ reads the hidden state of one model and adds a gated linear projection of that state into the residual stream of the other model, before the receiving model's next transformer layer processes it. This is structurally identical to the skip connections introduced by He et al. [5], except that the connection spans two independently trained models rather than two layers of the same model. No base model parameters are modified. Only the bridge projections and a small output mixing vector are trained.

The mechanism is tractable because independently trained transformers tend to develop structurally compatible internal geometries [8, 10, 19]. The relative positions of concepts in representation space are approximately preserved across models trained on different corpora, even when their coordinate systems differ. The difference between a generalist model's representation of a domain concept and a specialist model's representation of the same concept is therefore a learnable, structured quantity. The bridge learns to project this difference into a corrective update that the receiving model processes through its own frozen weights.

Bidirectional coupling adds a return bridge from the generalist to the specialist, creating a feedback loop in which each model's updated representation at layer ℓ becomes the source for the bridge running in the reverse direction. During training, the bridge gates learn to amplify components that produce consistent cross-model updates while suppressing components that do not. Because each model's training-specific noise is uncorrelated with the other model's representations, this mechanism has a regularising effect on the residual stream without any explicit supervision objective for it. Because both models execute in parallel and the bridges add only a matrix multiplication at each designated layer, the total inference time is determined by the slower of the two models rather than their sum.

We evaluate RC across four domains (medical, legal, coding, scientific) against MoE routing as the primary baseline. MoE is the natural point of comparison: it has access to the same two frozen models as RC and combines them without modifying either. We additionally report each model's individual perplexity under bridge influence and TruthfulQA Health accuracy to test whether fused-output improvements carry through to factual accuracy.

The paper is organized as follows. Section 2 covers related work. Section 3 describes the architecture. Section 4 provides theoretical motivation. Section 5 reports four experiments. Section 6 discusses deployment and use cases. Section 7 concludes.

The contributions of this paper are as follows:

1. Residual Coupling (RC): a modular architecture for connecting frozen language models through small learned latent bridge operators, without modifying any base model weights.
2. An analysis of how the linearity constraint in RC bridge projections distinguishes bridge learning from standard fine-tuning.
3. Empirical results across four domains showing that RC consistently outperforms MoE routing, reducing perplexity by up to 76% from the frozen baseline in the two-model setting and by 80.7% in the three-model setting, and improving TruthfulQA Health accuracy by up to 9 percentage points over the frozen baseline.
4. A theoretical account of why frozen models can coordinate through linear operators, grounded in representational convergence [8] and operational closure [16].

2. Related Work

Model stitching. The idea of connecting one model's intermediate representations to another's was introduced as a probe of representational similarity [3, 12]. A stitching layer, typically an affine map, is placed between the early layers of one frozen network and the later layers of another. Ainsworth et al. [1] address the related problem of permutation symmetry in independently trained models: their Git Re-Basin method shows that such models can often be aligned under permutation before weight merging, demonstrating that the inter-model geometry is non-trivial and structured. Stoica et al. [22] extend this to merging models trained on different tasks into a single network without further training. RC differs from both approaches in that it keeps the models fully separate and learns projection operators between their latent states during a parallel forward pass, without any alignment or merging step. Athanasiadis et al. [2] show that alignment at a single layer can miss distributional mismatches that emerge in subsequent layers, a concern that RC's multi-layer bridges address directly. RC also differs from standard stitching in being bidirectional, with coupling learned in both directions rather than imposed as a single cut point.

Representational convergence. Several lines of work document that neural networks trained on different objectives, modalities, and datasets develop increasingly aligned internal geometries as they scale [8, 10, 19]. Relative positions of concepts tend to be approximately preserved across models even when their coordinate systems differ. This convergence is what makes bridge training on frozen models tractable, since the representations are already structurally compatible before any bridge training begins.

Adapter-based fine-tuning. Adapter layers [6], low-rank adaptation [7], and prefix tuning [13] insert small trainable modules into a frozen backbone at a comparable parameter budget to RC's bridges. These methods are unidirectional and involve no second model, so improvements in factual accuracy require retraining the adapter on factual data rather than emerging as a structural consequence of coupling two independently trained models.

Cross-modal fusion. Attention bottleneck methods [18] and multimodal transformers [23] integrate features from different modalities through learned fusion layers. RC extends naturally to this setting, since a language model and a vision encoder, both frozen, could be coupled through bridge projections on their shared residual streams without architectural modification of either.

Mixture-of-Experts. MoE [4, 21] scales parameter counts efficiently through sparse routing but commits each token to a single expert. Prior work on multi-model systems reduces to selection (routing between models) or aggregation (merging their outputs or intermediate states). RC introduces a third approach: projection operators are learned between models' latent states, enabling structured mutual correction without selection or merging. RC's advantages over MoE are architectural rather than efficiency-based.

JEPA. JEPA [11] learns a world model by predicting latent representations of one view from another, operating entirely in embedding space. RC applies an analogous principle to correction rather than prediction: the bridges learn transformations between independently trained models rather than between views of a single model.

Universal embedding geometry. Jha et al. [9] show that large models develop structurally compatible representation spaces that support high-fidelity unsupervised translation. This is consistent with the convergence result discussed in Section 4.1 and provides additional support for the sufficiency of linear bridge projections.

3. Architecture

3.1 Transformers, Residual Streams, and Cross-Model Skip Connections

Transformer models are built around residual connections [5], sometimes called skip connections: each layer adds its output to the hidden state it received rather than replacing it. This additive structure means that the hidden state at any given layer accumulates contributions from all preceding layers and remains accessible as a continuous, high-dimensional vector at each sequence position.

RC exploits this structure directly. A bridge at layer ℓ computes a gated linear projection of one model's hidden state and adds it to the other model's hidden state before that model's next transformer layer processes it. This is structurally a skip connection [5], but one that spans two independently trained models rather than two layers of the same model. The receiving model processes the perturbed hidden state through its own frozen weights without any modification to those weights. No additional attention heads are introduced and no sequence positions are added. The computational overhead per bridge layer adds a matrix multiplication per designated layer. Since the two model stacks execute in parallel, inference latency scales with the depth of a single model, not with the number of coupled models.

3.2 Components

A unilateral bridge at layer ℓ from B to A is parameterized by $W^{(\ell, B\to A)} \in \mathbb{R}^{d_A \times d_B}$ and a scalar gate $g^{(\ell, B\to A)} \in \mathbb{R}$. It computes

$$\delta\mathbf{h}_A^{(\ell)} = \sigma(g^{(\ell, B\to A)})\, W^{(\ell, B\to A)}\, \mathbf{h}_B^{(\ell)} \tag{1}$$

$$\mathbf{h}_A^{(\ell)} \leftarrow \mathbf{h}_A^{(\ell)} + \delta\mathbf{h}_A^{(\ell)} \tag{2}$$

where σ is the sigmoid function. At initialization, $\sigma(-2) \approx 0.12$, so bridge contributions are initially small and increase only as permitted by the training signal. The bridge therefore learns to produce updates that are corrective with respect to the receiving model's current state, rather than degenerate identity mappings of the source features.

Bridge projections are implemented without bias terms. Layer normalization in the receiving model's subsequent layer implies that the perplexity difference between `bias=True` and `bias=False` is negligible across all experiments. A final learned mixing vector produces the output logits as a weighted sum over per-model logit tensors.

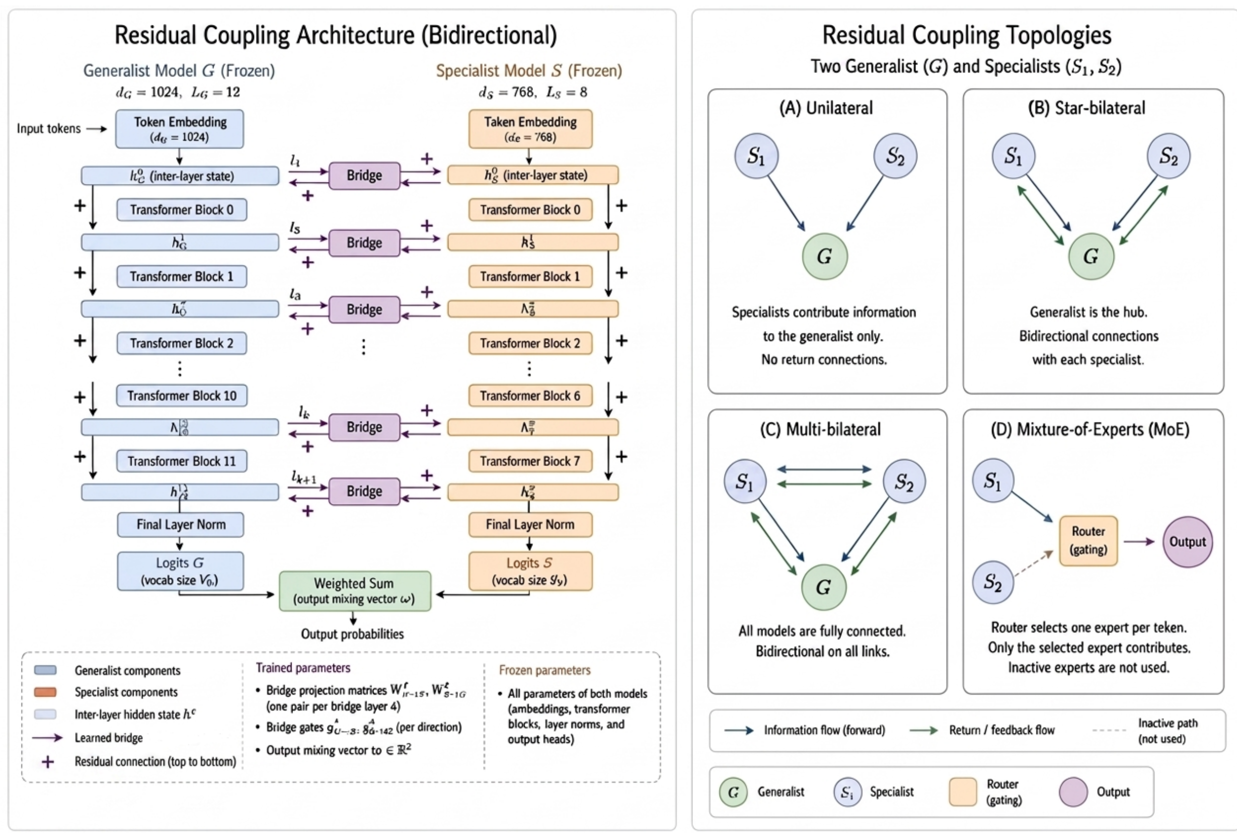

Figure 1: Left: RC architecture. **Right:** The four topologies.

3.3 Heterogeneous Model Pairs

Bridge projections handle dimensional mismatch naturally. When models A and B have different hidden dimensions d_A and d_B, the bridge $W_{B\to A}$ is a non-square linear map. Layer-count mismatch is resolved by proportional depth alignment: a bridge at generalist layer ℓ_A reads from specialist layer $\ell_B = \lfloor \ell_A \cdot L_B / L_A \rfloor$, where L_A and L_B are the respective number of layers. This permits coupling models of different sizes and depths without architectural modification of either. Experiment 1 in Section 5.1 uses this mechanism across four model pairs with different configurations.

3.4 Topologies

Unilateral. Each specialist injects into the generalist without reverse flow. The generalist is steered by the specialist but does not perturb it. This is the closest structural analogue to a unidirectional adapter.

Star-bilateral. The generalist and each specialist exchange bidirectional bridge updates, but specialists do not exchange with each other. The generalist acts as a central hub: each specialist pulls the generalist's representation toward its own domain knowledge, while the generalist's representations simultaneously provide corrective signal to each specialist.

Multi-bilateral. All pairs of models exchange bidirectional bridge updates. Bridge parameter count scales as $b \cdot n(n-1) \cdot d^2$, where n is the number of models and b is the number of bridge layers.

MoE (baseline). A learned router produces soft weights over the specialists' hidden states at each bridge layer, selecting among them rather than summing [4, 21]. This is the competitive baseline in all experiments.

3.5 Parameter Overhead

Table 1: Bridge parameter counts by topology. n = total number of models (generalist + specialists). For the three-model, $d = 768$, $b = 4$ configuration used in Experiments 2 and 3: approximately 4.7M, 9.4M, and 14.2M parameters for the three collaborative topologies respectively, against a negligible 2.3K for the MoE router and 124M per frozen base model. The multi-bilateral topology adds roughly 11% of a single base model's parameters per model pair. The frozen model weights are never copied or merged.

Topology	Bridge parameters
Unilateral	$b \cdot (n-1) \cdot d^2$
Star-bilateral	$b \cdot 2(n-1) \cdot d^2$
Multi-bilateral	$b \cdot n(n-1) \cdot d^2$
MoE router	$d \cdot n$ (negligible)

4. Theoretical Motivation

4.1 The Geometry of Latent Coordination

Several independent lines of work have documented a relevant empirical regularity: independently trained transformer models tend to develop structurally compatible internal geometries [8, 10, 19]. Relative distances between concepts are approximately preserved across models even when their coordinate systems differ. This compatibility is what makes linear bridge projections sufficient to transfer information usefully between frozen models without any shared pretraining or alignment procedure.

The sufficiency of a linear projection between nonlinear manifolds is supported by converging empirical evidence. Linear probes applied to frozen hidden states recover semantic structure reliably across diverse tasks, indicating that the latent manifolds of trained transformers are locally linear in semantically relevant directions. Procrustes alignment and Canonical Correlation Analysis studies [8, 19] show that the principal axes of variation in one model's representation space map onto an other's with low residual error, particularly in deeper layers. Jha et al. [9] extend this finding to large-model embedding spaces, demonstrating high-fidelity unsupervised translation using only linear maps. The shared causal structure that independently trained models encode appears to produce corridors of approximately linear geometry between their manifolds, and it is these corridors that the bridge projections navigate.

Huh et al. [8] interpret this convergence as models moving toward a shared statistical model of the world, a geometry encoding the causal structure of reality rather than the particulars of any training corpus. Under this view, the bridge does not memorize noise but navigates a pre-existing geometric relationship between independent manifolds. A corollary is that the magnitude of the bridge's contribution should scale with the distributional distance between the two models: when the specialist's training distribution is close to the generalist's, the representational gap is smaller and the bridge has less corrective work to do. This is consistent with the modest gains observed in the scientific domain in Experiment 1, where the two models are trained on overlapping material.

Each trained model's outputs combine shared signal with private noise. The shared signal is the factual and linguistic structure that any large and varied training corpus encodes. The private noise is model-specific confabulation arising from gaps, biases, and distributional artifacts in that model's particular training corpus. Private noise is statistically uncorrelated across models trained on different corpora. During training, the bridge gates learn to amplify projected components that produce consistent updates across models while suppressing components that do not. The TruthfulQA results in Section 5.3 are consistent with this mechanism: factual accuracy improves alongside perplexity reduction across all RC topologies. Hallucination arises partly from this private noise: a model produces confident but false outputs when its training distribution contains gaps that its internal geometry fills by interpolation rather than by memorized fact. Cross-model gating suppresses components of this type by preferring updates that are consistent across independently trained models.

The bridge is linear and therefore it can only learn continuous linear relationships between the representation spaces the frozen models define. A non-linear bridge would recover expressive power but lose this guarantee: it could memorize, and the source of any gain would become difficult to interpret. We treat the linearity as a design choice rather than a limitation. The frozen transformers act as non-linear memorizers that define the boundaries of the latent manifold, while the bridges are linear generalization spaces that navigate the structured differences between them.

4.2 Frozen Models, Operational Closure, and Relational Computation

When a model is fine-tuned, its parameters are modified in response to new data. The system that emerges from training is not the same system that entered it, and the internal organisation that supported prior capabilities may be partially or fully overwritten. This is the structural cause of catastrophic forgetting. RC avoids this by design: all models are frozen before any bridge training begins, and their weights remain invariant throughout. Each frozen model receives bridge injections into its residual stream at designated layers, but those injections do not modify the receiving model's weights. The injected signal is assimilated through the model's existing computational structure and does not alter it. In Maturana and Varela's [16] terms, this is operational closure: the model does not change to accommodate its input but transforms its input according to its own internal organization. Catastrophic forgetting is not prevented by regularization here. It is made structurally impossible because the condition that produces it is never present.

The architecture follows a last-in-first-out protocol for module addition and removal. Because each new specialist is trained on the manifold established by the existing ensemble, deactivating modules in reverse-chronological order ensures that the remaining bridges navigate a representation space consistent with their training conditions.

During training, bridge projections learn to suppress components that are already well-represented in the target model and to amplify components that provide corrective information. Within a single transformer, representations arise from compositions of attention and non-linear activations, producing a piecewise-linear partitioning of representation space. The linear bridge operators map between independently learned manifolds, enabling corrective trajectories between representation spaces rather than within any single one.

5. Experiments

All experiments use a causal language modelling objective with gradient accumulation over 8 steps. All runs use seed 42 with deterministic CUDA settings. Perplexity is reported relative to the frozen generalist anchor (Baseline A). MoE absolute perplexity is included as a reference for the competitive routing alternative.

5.1 Experiment 1: Domain Generality

We test whether RC generalizes across domains, model sizes, and training distributions. We pair a GPT-2 generalist anchor with a domain specialist in four configurations and compare bilateral coupling against the two frozen baselines, MoE routing, and unilateral coupling. The question is not whether RC beats the frozen models in isolation, but whether it beats MoE, which also has access to both models and represents the strongest available approach for combining them without modifying either.

GPT-2 [20] serves as the generalist anchor in all four configurations. Bridge layers are selected proportionally to model depth: every sixth layer for 36-layer models, every fourth for 24-layer models, every third for 12-layer models.

The MoE baseline uses the same frozen model pair as RC, trained for the same number of steps (`MAX_STEPS`: 2,000) with the same optimizer and learning rate schedule. The router is a single linear layer producing soft weights over the specialists' hidden states at each bridge layer, totaling approximately 2.3K parameters against 4.7M to 14.2M for RC bridges. Experiment 4 compares MoE against random-projection RC to confirm that the gains require learned projection structure, not a larger parameter count.

Table 2: Perplexity by domain and topology (all fused outputs). Bilateral coupling outperforms MoE in all four domains. `MAX_STEPS`: 2,000. `TEST_SAMPLES`: 25. Seed: 42. Source: `benchmark.py`.

Domain	Generalist	Specialist	Fr.A	Fr.B	MoE	Uni	Bi
Medical	GPT-2 Med. (345M)	DialoGPT-Med. (345M)	50.05	317.89	64.66	13.08	**12.01**
Scientific	GPT-2 Lg. (774M)	gpt2-large-medical (774M)	28.54	28.77	26.85	18.73	**17.51**
Coding	GPT-2 (124M)	CodeGPT-small-py (124M)	16.68	∼7M	878.40	15.15	**5.91**
Legal	GPT-2 (124M)	open-australian-legal-gpt2	26.48	44.44	21.83	9.50	**8.30**

Bilateral coupling outperforms both the frozen models and MoE routing in all four domains. The coding result is the most informative. CodeGPT-small-py uses a distinct tokenizer from GPT-2, producing a frozen perplexity of approximately 7 million on the evaluation text. Surface-level aggregation fails entirely: MoE reaches 878.40 and logit ensemble reaches 596.16, both substantially worse than the frozen generalist alone (16.68). Bilateral coupling reaches 5.91. The bridge gates learn, without any explicit vocabulary alignment procedure, to extract latent representations that carry useful corrective signal from a model whose output distribution is incompatible with the evaluation domain. This is possible because the bridge operates on hidden states rather than on logits.

Unilateral coupling (15.15) achieves a modest fused-output improvement over the frozen generalist (16.68) in the coding domain, but bilateral coupling reaches 5.91. The coding domain also illustrates why bidirectionality matters beyond fused-output metrics: under unilateral coupling, the generalist's individual steered output degrades to 32.11, considerably worse than its frozen baseline. The training objective optimizes the fused output at the expense of the generalist's own residual stream when it lacks a return signal. The reverse bridge corrects this: under bilateral coupling, the generalist's individual output recovers to 11.29. The scientific domain shows the smallest gains overall, consistent with the specialist already performing close to the generalist on the evaluation domain.

Table 3: Steered agent perplexity for unilateral and bilateral topologies. Steered A: model A's individual output after bridge influence. Steered B: model B's individual output after bridge influence. Domain gain is reduction from Baseline A. In unilateral coupling, Steered B is unchanged from its frozen baseline, as model B receives no bridge signal from model A. The coding Steered B remains high under bilateral coupling because the tokenizer mismatch prevents coherent general-text generation from model B individually. The fused output succeeds (5.91) because the bridge extracts useful latent signal without relying on model B's token-level output. Source: `benchmark.py`.

Domain	Topology	Fused PPL	Steered A	Steered B	Domain gain
Medical	Unilateral	13.08	23.62	317.89 (frozen)	+73.87%
Medical	Bilateral	12.01	15.54	22.59	+76.01%
Legal	Unilateral	9.50	11.71	44.44 (frozen)	+64.11%
Legal	Bilateral	8.30	10.64	10.84	+68.65%
Scientific	Unilateral	18.73	19.77	28.77 (frozen)	+34.39%
Scientific	Bilateral	17.51	18.35	18.24	+38.66%
Coding	Unilateral	15.15	32.11	7,010,479 (frozen)	+9.22%
Coding	Bilateral	5.91	11.29	19,461	+64.60%

The steered agent results show that in bilateral coupling, both models improve individually, not only through the fused output. In the medical domain, Steered B drops from 317.89 to 22.59. In the legal domain it drops from 44.44 to 10.84. In the scientific domain, Steered A (18.35) and Steered B (18.24) converge to near-identical perplexity despite having been trained on different corpora. This bidirectional improvement follows from the feedback loop: each model's updated residual stream continuously steers the other's representations, and both streams benefit.

5.2 Experiment 2: Multi-Specialist Topology Sweep

We scale to three frozen models in the medical domain: GPT-2 (124M) as the generalist anchor, DialoGPT-small and a fine-tuned medical QA model as specialists. We test all four primary topologies against MoE and the frozen baseline. As shown in Table 4, all RC topologies reduce perplexity by approximately 80% relative to the frozen baseline, while MoE achieves 0.5%. The PPL differences between the three collaborative RC topologies are small (11.02 to 11.26), though a single domain, dataset, and distribution is insufficient to generalize from.

Table 4: Multi-specialist topology sweep, medical domain, three frozen models. Baseline A: GPT-2 (PPL 57.08). Hybrid: star-bilateral topology with per-layer entropy-weighted mixing. Linear bridge projection. `MAX_STEPS`: 2,000. `TEST_SAMPLES`: 50. Seed: 42. Source: `three_qa.py`.

Topology	PPL	Gain from Baseline A
Multi-Bilateral	**11.02**	+80.7%
Hybrid	11.11	+80.5%
Star-Bilateral	11.07	+80.6%
Multi-Unilateral	11.26	+80.3%
MoE	56.80	+0.5%
Baseline A (frozen)	57.08	reference

5.3 Experiment 3: Factual Accuracy

Perplexity measures language modeling quality but not factual accuracy, and improvements in perplexity can in principle coexist with increased hallucination if a model becomes more fluent at producing false statements. We evaluate whether the perplexity improvements in Experiment 2 carry over to verifiable factual claims using TruthfulQA Health [14], a set of medical questions with one correct answer and several plausible alternatives. Each model scores each choice by the mean log-probability of its tokens conditioned on the question and selects the highest-scoring choice. No generation and no model-specific prompting are required.

Table 5: TruthfulQA Health (MC1), medical domain. PPL and accuracy across topologies. TQA gain in percentage points relative to Baseline A (16.36%). Source: `three_qa.py`.

Topology	PPL	TruthfulQA (%)	TQA gain (pp)
Multi-Bilateral	11.02	**25.45**	+9.09
Multi-Unilateral	11.26	23.64	+7.28
Hybrid	11.11	23.64	+7.28
Star-Bilateral	11.07	21.82	+5.46
MoE	56.80	20.00	+3.64
Baseline A (frozen)	57.08	16.36	reference

Multi-bilateral achieves the highest TruthfulQA accuracy (25.45%), followed by multi-unilateral and hybrid at 23.64% and star-bilateral at 21.82%. MoE reaches 20.00%, only marginally above the frozen baseline of 16.36%. Factual accuracy is not strictly ordered by topology complexity: multi-unilateral matches the hybrid despite simpler coupling. This suggests that, for these questions, the primary factor is whether the specialist's representations reach the generalist, not whether the correction is bidirectional. All RC topologies improve factual accuracy relative to both the frozen baseline and MoE. The consistent improvement in factual accuracy across all RC topologies suggests that cross-model correction has a measurable effect on hallucination, not only on language modeling quality.

5.4 Experiment 4: Ablation

We run an ablation across all four domains from Experiment 1 to test two questions: whether the gains require learned projection structure rather than merely the architectural position of the bridge, and whether bidirectionality contributes beyond unidirectional injection. We test six conditions: logit ensemble, MoE, unilateral, bilateral, bilateral without learned gates, and bilateral with random fixed projections (gates trained, projections frozen at random initialization). All other hyperparameters and model pairs are identical to Experiment 1.

The random projection condition is the most informative in the ablation. In the medical domain, random bridges produce a fused PPL of 109.55 against a frozen baseline of 50.05. In the coding domain the result is 499.93. In the three-model setting (`three_qa.py`), `multi_bilateral_random` reaches 166.82 against a frozen baseline of 57.08, while trained multi-bilateral reaches 11.02. The bridge's gains therefore require learned projection structure and not merely the presence of an additive connection or a trained gate.

The logit ensemble fails in every domain with a large distributional gap. In the coding domain it reaches 596.16 against a frozen baseline of 16.68; in the medical domain 69.49 against 50.05. When one model is far out of distribution, averaging logits at the output amplifies its incoherence rather than canceling it. The bridge's advantage is precisely that it operates on hidden states, where the structure learned by the out-of-distribution model remains useful even when its token-level outputs are not.

The gate ablation gives a nuanced result. Bilateral without learned gates marginally outperforms gated bilateral in the legal domain (8.13 vs. 8.30) and the coding domain (4.95 vs. 5.91), while gated bilateral wins in the medical and scientific domains. The gate is therefore not universally beneficial. It is a conservative default that prevents early-training instability in domains where the bridge has not yet learned a useful differential, at a small cost in cases where the corrective signal is reliable from the first training steps. In the three-model setting the pattern reverses: no-gate reaches 16.42 against multi-bilateral's 11.02, suggesting that the gate's stabilizing effect becomes more important as the number of coupled models increases.

In the multi-specialist setting (`three_qa.py`), PPL differences between topologies are small: multi-bilateral reaches 11.02, star-bilateral 11.07, multi-unilateral 11.26. In the two-model coding domain, bidirectionality has a more decisive effect: unilateral coupling achieves a fused PPL of 15.15 but degrades the generalist's individual output to 32.11. Bilateral coupling recovers the generalist's individual output to 11.29 and achieves a fused PPL of 5.91. When the distributional gap between models is large, the reverse bridge stabilizes the generalist's residual stream. Without it, the training objective for the fused output corrupts the generalist's individual representations.

We also compare single linear projection against a bottleneck variant ($d \to d/2 \to d$). The perplexity difference is negligible across all conditions. We retain single linear projection as the default for architectural simplicity.

Table 6: Ablation results (perplexity, lower is better). Baseline A is the frozen generalist for each domain. Random: projection matrices fixed at random initialization, only gate values trained. *In coding, bilateral no-gate (4.95) narrowly outperforms gated bilateral (5.91); see text. `MAX_STEPS`: 2,000. `TEST_SAMPLES`: 25. Seed: 42. Source: `benchmark.py`.

Domain	Logit ens.	MoE	Uni	Bi	No gate	Random	Baseline A
Medical	69.49	64.66	13.08	**12.01**	13.73	109.55	50.05
Scientific	27.56	26.85	18.73	**17.51**	21.44	28.54	28.54
Coding	596.16	878.40	15.15	5.91	**4.95***	499.93	16.68
Legal	25.33	21.83	9.50	8.30	**8.13**	26.11	26.48

6. Deployment and Use Cases

RC occupies a position at the intersection of three families of methods that currently dominate applied language modeling: parameter-efficient fine-tuning, routing-based multi-model systems, and multi-agent pipelines. It shares properties with each while avoiding some of their limitations.

Parameter-efficient fine-tuning methods such as LoRA [7] insert small trainable modules into a frozen or partially frozen backbone at a parameter overhead comparable to RC's bridges. The fundamental difference is that LoRA modifies the base model's effective weights, which means it inherits the forgetting properties of standard fine-tuning. Liu et al. [15] demonstrate this directly: finetuning on a naturally occurring writing task causes models to reproduce verbatim passages from their training corpora, indicating that fine-tuning reactivates latent memorization from pretraining as a side effect of learning the new task. RC trains only on the gap between two models rather than on the weights of either, leaving all base model parameters untouched.

MoE routing selects among specialists per token but does not allow specialists to influence each other's intermediate computations. In the three-model medical setting, MoE with access to all three models achieves a fused perplexity of 56.80 against the frozen baseline of 57.08, a reduction of 0.5%. Multi-bilateral coupling achieves 11.02. Routing selects among specialists but prevents the continuous cross-model correction that RC enables at the level of hidden states.

Multi-agent pipelines coordinate models through text, compressing each model's continuous intermediate representations into a discrete token sequence at every handoff. RC operates on hidden states throughout a single parallel forward pass. For tasks where the relevant signal is geometric, meaning it depends on where a concept sits in representation space rather than on how it is verbally expressed, operating on hidden states rather than token sequences is an advantage. The coding experiment makes this concrete: CodeGPT-small-py is entirely out of distribution on general text, with a frozen perplexity of approximately 7 million. Any method that operates on its output logits fails. RC reaches a fused perplexity of 5.91 by reading the specialist's hidden states directly.

Table 7 summarizes the comparison across these three families.

Table 7: Architectural comparison across specialization strategies.

Property	**LoRA**	**MoE**	**Symbolic agents**	**RC**
Base model modified	Yes	No	No	No
Forgetting risk	Present	None	None	None (structural)
Specialist integration	Requires retraining	Requires retraining	Modular	Bridge training for new module
Communication medium	N/A	Token routing	Discrete tokens	Continuous latent vectors
Inference latency	Single pass	Single pass + router	Sequential multi-pass	Parallel pass, depth of single model
Bridge parameter overhead	Low	Negligible	None	Low to moderate
Factual correction	None	Competitive routing	Iterative prompting	Per-layer collaborative injection

The practical implication is that RC is applicable today, on existing hardware, to problems currently being addressed by fine-tuning or routing. An organization with a general-purpose base model and a domain specialist can couple them through bridge training without modifying either model. At $d = 768$ with four bridge layers, the bilateral bridge between two models is approximately 4.7M parameters, around 2% of two 124M parameter models (total 248M frozen parameters). The bridge parameters contain no training data from either base model and can be stored, versioned, and transferred independently of the model weights.

This opens a deployment pattern that fine-tuning and routing cannot support. A specialist model could in principle run on a local or edge device while the generalist anchor runs on a remote server, with input tokens and intermediate hidden states crossing the network boundary at each bridge layer, though this deployment pattern has not been tested. Neither model's weights are exposed to the other party. The steered agent results from Experiment 1 show that bilateral coupling improves the specialist's individual output, not only the fused output, so the edge device can generate improved responses independently while coupled.

The model lifecycle problem is also more tractable under RC than under fine-tuning. Specialists and their bridges are added sequentially, each trained on the manifold established by the existing ensemble. When a domain shifts and a specialist needs to be updated, the old specialist is frozen in place and a new one is trained and bridged. Because each addition is independent of prior additions at the weight level, a specialist can be removed by deactivating its bridges in the reverse order they were added, leaving all prior components untouched. The system reverts exactly to its prior state without any retraining. This is more straightforwardly reversible than LoRA rank switching and more auditable than MoE router retraining.

The experiments reported here use models in the 124M to 774M parameter range with a single domain per specialist. The behavior of RC at larger scales, with larger specialist ensembles, and with models trained from the start with coupling in mind, is a direction for future work.

The three-model results raise a scaling question. Multi-bilateral coupling at $n = 3$ reduces perplexity by 80.7% against MoE's 0.5%, and bridge parameter count grows as $O(n^2)$ while inference latency does not, since parallel execution keeps it time bounded by the depth of the slowest model regardless of n. If the representational convergence hypothesis [8] holds at larger scales, the corrective signal from each additional specialist should grow with model size even as the geometric distance between models narrows, producing a more structured bridge target. An ensemble of ten or more frozen specialists, each trained on a distinct domain or modality and connected through RC bridges, would carry a total parameter count dominated by the frozen columns, with bridge training cost scaling in specialist pairs. Whether the horizontal scaling axis produces capability gains qualitatively different from those observed at $n = 3$ remains to be tested.

An agent workflow with two models incurs at minimum two sequential forward passes plus the latency of the text handoff between them. RC with two coupled models incurs one forward pass at the cost of the bridge projections, which add $O(d^2)$ computation at each bridge layer.

The bridge mechanism is not modality-specific. A language model and a vision encoder with compatible residual stream dimensions could in principle be coupled through the same projection structure, and the same logic extends to pairing language models with time-series or forecasting models, where the specialist encodes distributional patterns over structured sequences rather than text. These are directions for future work.

7. Conclusion

> "The pure present is an ungraspable advance of the past devouring the future. In truth, all sensation is already memory."
>
> *Henri Bergson, Matter and Memory* (1896)[2]

We have presented Residual Coupling, a method for connecting frozen language models through gated linear bridge projections that add corrective updates into each model's residual stream. The bridge is structurally a cross-model skip connection [5]: the same additive mechanism that residual networks use within a single model, extended to span two independently trained ones. No base model weights are modified at any point, so base model representations are immune to catastrophic forgetting by construction. Because the coupled models execute in parallel, inference latency is determined by the depth of the slower model, not by the number of specialists in the ensemble. Where current scaling practice adds capacity by increasing model depth, RC adds capacity by training lightweight connections between models of fixed depth, a horizontal scaling axis that leaves each component independently interpretable and replaceable.

Across four domains, RC consistently outperforms MoE routing with the same frozen models. In the three-model medical setting, multi-bilateral coupling reduces perplexity by 80.7% relative to the frozen baseline, compared to 0.5% for MoE, and achieves 25.45% accuracy on TruthfulQA Health against 20.00% for MoE and 16.36% for the frozen baseline. Adding a new specialist requires training bridges to a new frozen module, typically in a few thousand steps on a single GPU. Because specialists and their bridges are added sequentially, a specialist can be removed by deactivating its bridges in reverse order of addition. The remaining components are untouched and the system reverts to its prior state without retraining.

Two properties of the bridge account for its effectiveness. Operating on hidden states rather than on logits allows it to extract useful signal from a model whose token-level output is incompatible with the evaluation domain, as in the coding experiment where the specialist's frozen perplexity on general text is approximately 7 million. The linearity constraint means the bridge learns a continuous map between the memorized spaces defined by the frozen models. We view this as a desirable property rather than a limitation, since it makes the bridge's function interpretable and its generalization capacity traceable to the relational geometry between models.

The theoretical account in Section 4 draws on two bodies of work. Representational convergence [8, 10] explains why bridge training on frozen models is tractable: as models scale,

[2] Quoted by a character in Haruki Murakami's *Kafka on the Shore* (2002), consistent with the verbatim recall of Murakami's works documented by Liu et al. [15].

their internal geometries converge toward a shared structure, so the difference between two models' representations is navigable by a linear operator. Operational closure [16] explains why the frozen design is not merely convenient but correct: a frozen model maintains its identity across the lifetime of the coupled system, processing each bridge injection through its own unchanged weights without being modified by it.

If the approach scales, an iterative training approach becomes natural. In the first stage, specialist columns are trained independently and frozen, each accumulating domain knowledge without exposure to the others. In the second stage, bridges are trained on the frozen ensemble, learning the differential operators that coordinate what the columns have separately encoded. Mountcastle [17] proposed that the neocortex follows an analogous organization, with functionally specialized columns operating independently yet collectively producing unified perception through their lateral interactions. Whether this modular growth principle extends to larger models and across modalities remains to be tested.

Data and Code Availability

All datasets used in this study are publicly available. Implementation and trained model configurations are provided at: https://github.com/pfekin/residual-coupling

References

[1] Ainsworth, S., Hayase, J., and Srinivasa, S. (2022). Git re-basin: Merging models modulo permutation symmetries. *arXiv*:2209.04836.

[2] Athanasiadis, I., et al. (2025). Model stitching by functional latent alignment. Preprint.

[3] Bansal, Y., Nakkiran, P., and Barak, B. (2021). Revisiting model stitching to compare neural representations. *Advances in Neural Information Processing Systems*, 34.

[4] Fedus, W., Zoph, B., and Shazeer, N. (2022). Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120).

[5] He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

[6] Houlsby, N., Giurgiu, A., Jastrzebski, S., et al. (2019). Parameter-efficient transfer learning for NLP. *Proceedings of the 36th International Conference on Machine Learning*.

[7] Hu, E. J., Shen, Y., Wallis, P., et al. (2021). LoRA: Low-rank adaptation of large language models. *arXiv*:2106.09685.

[8] Huh, M., Cheung, B., Wang, T., and Isola, P. (2024). The Platonic Representation Hypothesis. *arXiv*:2405.07987.

[9] Jha, R., Zhang, C., Shmatikov, V., and Morris, J. X. (2025). Harnessing the universal geometry of embeddings. *arXiv*:2505.12540.

[10] Kornblith, S., Norouzi, M., Lee, H., and Hinton, G. (2019). Similarity of neural network representations revisited. *Proceedings of the 36th International Conference on Machine Learning.*

[11] LeCun, Y. (2022). A path towards autonomous machine intelligence. OpenReview preprint.

[12] Lenc, K. and Vedaldi, A. (2015). Understanding image representations by measuring their equivariance and equivalence. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.*

[13] Li, X. L. and Liang, P. (2021). Prefix-tuning: Optimizing continuous prompts for generation. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics.*

[14] Lin, S., Hilton, J., and Evans, O. (2022). TruthfulQA: Measuring how models mimic human falsehoods. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics.*

[15] Liu, X., Mireshghallah, N., Ginsburg, J. C., and Chakrabarty, T. (2026). Alignment whack-a-mole: finetuning activates verbatim recall of copyrighted books in large language models. *arXiv*:2603.20957.

[16] Maturana, H. and Varela, F. (1980). *Autopoiesis and Cognition: The Realization of the Living.* Reidel.

[17] Mountcastle, V. B. (1978). An organizing principle for cerebral function: The unit module and the distributed system. In Edelman, G. M. and Mountcastle, V. B. (Eds.), *The Mindful Brain.* MIT Press.

[18] Nagrani, A., Yang, S., Zisserman, A., et al. (2021). Attention bottlenecks for multimodal fusion. *Advances in Neural Information Processing Systems*, 34.

[19] Nguyen, T., Raghu, M., and Kornblith, S. (2021). Do wide and deep networks learn the same things? *Proceedings of the 9th International Conference on Learning Representations.*

[20] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI Blog.*

[21] Shazeer, N., Mirhoseini, A., Maziarz, K., et al. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv*:1701.06538.

[22] Stoica, G., Bolya, D., Bjorner, J., et al. (2023). ZipIt! Merging models from different tasks without training. *arXiv*:2305.03053.

[23] Tsai, Y.-H. H., Bai, S., Liang, P. P., et al. (2019). Multimodal transformer for unaligned multimodal language sequences. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

A. Experimental Configuration

A.1 Script Descriptions

`benchmark.py` (Experiments 1 and 4). One generalist, one specialist, six conditions (logit ensemble, MoE, unilateral, bilateral, bilateral without learned gates, bilateral with random fixed projections), four domains. Each domain uses the appropriate model pair and bridge layer spacing for the model depth, covering proportional depth alignment for heterogeneous model pairs. The script reports fused perplexity, Steered A, and Steered B for every condition. Results are embedded as comments at the end of the file.

`three_qa.py` (Experiments 2 and 3). One generalist, two specialist modules, eight conditions (logit ensemble, multi-unilateral, star-bilateral, multi-bilateral, hybrid multi-bilateral, multi-bilateral without gates, multi-bilateral with random projections, MoE), medical domain. Includes TruthfulQA Health evaluation under the MC1 protocol for all three model outputs. As in `benchmark.py`, results are embedded as comments at the end of the file.

A.2 Implementation Notes

Vocabulary alignment across heterogeneous model pairs is handled by `torch.clamp`, clamping token indices to the smaller vocabulary size before each model's embedding lookup. Bridge projections are implemented without bias terms (`bias=False`). The perplexity difference between `bias=True` and `bias=False` is negligible across all experiments.

A.3 Hyperparameters

Table 8: Hyperparameters used across all experiments. Gradient accumulation over 8 steps is used in all runs to keep GPU memory requirements within single-card constraints; peak VRAM usage is approximately 9 GB in both scripts. The effective batch size is identical to a direct batch of the same total size.

Parameter	`benchmark.py`	`three_qa.py`
`MAX_STEPS`	2,000	2,000
`GRAD_ACCUM`	8	8
`MAX_SEQ_LEN`	128	128
`TEST_SAMPLES`	25	50
Bridge learning rate	1e-4	1e-4
Router / mixing learning rate	5e-3	5e-3
Gate initialization	-2.0	-2.0
Optimizer	AdamW	AdamW